

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		104508	
Total Threats Mitigated	50	Total Packets Analyzed		5536544109	
High Priority	17	Malicious Packets		5516892344	
Low Priority	33	True Negatives Passed		83842	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		6	
ML Metrics	Value	SYN Flood		18	
False Positives	249	UDP Flood		12	
FP Rate	0.23%	Uncertain		14	
IF Precision	99.01%				
IF Recall	94.15%	Action		Count	
IF F1-Score	96.52%	Quarantined		11	
IF Accuracy	98.66%	Time Ban		10	
		Blackhole		0	
		Released		8	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.01%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.15%	Share of actual attacks successfully flagged
F1-Score	96.52%	Balanced measure combining Precision and Recall
Accuracy	98.66%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.23%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.85%	Actual attacks that went undetected
True Positive Rate (TPR)	94.15%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.77%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 19092	FN 1187
Actual: Normal	FP 191	TN 82655

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.96%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.98%	Balanced measure combining Precision and Recall
Accuracy	99.96%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	3.66%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	96.34%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	9433	0	0
Act: ICMP	0	2825	0
Act: UDP	0	0	6834

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.01%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.15%	Share of actual attacks successfully flagged
F1-Score	96.52%	Balanced measure combining Precision and Recall
Accuracy	98.66%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.23%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.85%	Actual attacks that went undetected
True Positive Rate (TPR)	94.15%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.77%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 19092	FN 1187
Actual: Normal	FP 191	TN 82655

2d. Response Latency

Metric	Value	Description
Detection Time	413.3 ms	Average time taken to detect an attack
Mitigation Response Time	10.0 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	2.70%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	10.80%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	4.74%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.17	3	3	2026-09-25 12:20:50	2026-09-25 12:24:01
10.0.0.18	3	3	2026-09-25 12:24:06	2026-09-25 12:25:28
10.0.0.8	3	3	2026-09-25 12:22:57	2026-09-25 12:25:17
10.0.0.20	2	3	2026-09-25 12:20:50	2026-09-25 12:23:51
10.0.0.16	2	2	2026-09-25 12:24:10	2026-09-25 12:25:20
10.0.0.19	2	2	2026-09-25 12:24:07	2026-09-25 12:25:18
10.0.0.21	1	1	2026-09-25 12:24:37	2026-09-25 12:24:47
10.0.0.22	1	1	2026-09-25 12:24:37	2026-09-25 12:24:48

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.24	1	1	2026-09-25 12:25:09	2026-09-25 12:25:20
10.0.0.25	1	1	2026-09-25 12:25:07	2026-09-25 12:25:18
10.0.0.1	3	1	2026-09-25 12:22:52	2026-09-25 12:23:24
10.0.0.9	1	1	2026-09-25 12:24:04	2026-09-25 12:24:15

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 12:06:02	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:02	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:03	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:03	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:12	10.0.0.18	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-25 12:06:12	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-25 12:06:12	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-25 12:06:12	10.0.0.18	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-25 12:06:13	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-25 12:06:13	10.0.0.19	DDoS	SYN Flood	92.5%	Critical	Time Ban (1m)
2026-09-25 12:06:13	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-25 12:06:13	10.0.0.19	DDoS	SYN Flood	92.5%	Critical	Time Ban
2026-09-25 12:06:33	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:33	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:34	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:43	10.0.0.20	DDoS	UDP Flood	75.6%	High	Time Ban (1m)
2026-09-25 12:06:43	10.0.0.20	DDoS	UDP Flood	75.6%	Critical	Time Ban
2026-09-25 12:06:44	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban (1m)
2026-09-25 12:06:44	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban (1m)
2026-09-25 12:06:44	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban
2026-09-25 12:06:44	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban
2026-09-25 12:07:03	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:03	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:05	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:12	10.0.0.18	DDoS	SYN Flood	92.7%	High	Released
2026-09-25 12:07:12	10.0.0.16	DDoS	SYN Flood	92.7%	High	Released
2026-09-25 12:07:13	10.0.0.17	DDoS	SYN Flood	93.3%	High	Time Ban (2m)
2026-09-25 12:07:13	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Released
2026-09-25 12:07:13	10.0.0.23	DDoS	ICMP Flood	90.8%	High	Time Ban (1m)
2026-09-25 12:07:14	10.0.0.23	DDoS	ICMP Flood	90.8%	Critical	Time Ban
2026-09-25 12:07:14	10.0.0.25	DDoS	ICMP Flood	91.2%	High	Time Ban (1m)
2026-09-25 12:07:15	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban
2026-09-25 12:07:15	10.0.0.24	DDoS	ICMP Flood	91.2%	Critical	Time Ban (1m)
2026-09-25 12:07:16	10.0.0.24	DDoS	ICMP Flood	91.2%	High	Time Ban
2026-09-25 12:07:23	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)
2026-09-25 12:09:58	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-25 12:11:58	10.0.0.19	DDoS	SYN Flood	92.7%	Critical	Time Ban (5m)

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 12:11:58	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (5m)
2026-09-25 12:11:58	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Time Ban (5m)
2026-09-25 12:13:02	10.0.0.4	DDoS	Uncertain	59.6%	Low	Released
2026-09-25 12:13:02	10.0.0.9	DDoS	UDP Flood	92.0%	High	Time Ban (1m)
2026-09-25 12:13:08	10.0.0.10	DDoS	Uncertain	68.4%	Low	Released
2026-09-25 12:13:15	10.0.0.16	DDoS	SYN Flood	92.5%	Critical	Time Ban (2m)
2026-09-25 12:21:51	10.0.0.20	DDoS	UDP Flood	79.4%	High	Time Ban (2m)
2026-09-25 12:22:57	10.0.0.8	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-25 12:23:03	10.0.0.1	DDoS	Uncertain	58.9%	Low	Released
2026-09-25 12:23:51	10.0.0.20	DDoS	UDP Flood	79.4%	High	Time Ban (5m)
2026-09-25 12:23:58	10.0.0.8	DDoS	UDP Flood	92.4%	High	Released
2026-09-25 12:24:15	10.0.0.9	DDoS	Uncertain	48.4%	Low	Released
2026-09-25 12:24:26	10.0.0.8	DDoS	UDP Flood	92.4%	High	Quarantined

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.18	SYN Flood	0.6381	94.0%	Critical	Phase 2	3	10s	2026-09-25 12:25:28	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6350	94.2%	Critical	Phase 2	2	1m 10s	2026-09-25 12:25:20	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6460	91.2%	Critical	Phase 2	1	11s	2026-09-25 12:25:20	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6424	93.7%	Critical	Phase 2	2	1m 11s	2026-09-25 12:25:18	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6461	91.0%	High	Phase 2	1	11s	2026-09-25 12:25:18	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6062	94.0%	Critical	Phase 2	1	1m 11s	2026-09-25 12:25:17	Ban Expired
10.0.0.8	UDP Flood	0.6126	92.4%	High	Phase 1	3	11s	2026-09-25 12:25:17	Attack Stopped
10.0.0.22	UDP Flood	0.6367	86.4%	High	Phase 2	1	11s	2026-09-25 12:24:48	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6300	79.6%	High	Phase 2	1	10s	2026-09-25 12:24:47	Attack persisted - escalating to time ban
10.0.0.8	UDP Flood	0.6219	92.4%	High	Phase 1	2	11s	2026-09-25 12:24:37	Attack Stopped
10.0.0.16	SYN Flood	0.6451	94.2%	Critical	Phase 2	1	10s	2026-09-25 12:24:20	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6501	93.7%	Critical	Phase 2	1	11s	2026-09-25 12:24:18	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6464	92.2%	Critical	Phase 2	1	11s	2026-09-25 12:24:17	Attack persisted - escalating to time ban
10.0.0.9	Uncertain	0.6176	48.4%	Low	Phase 1	1	11s	2026-09-25 12:24:15	Attack Stopped
10.0.0.17	SYN Flood	0.6466	92.2%	Critical	Phase 2	3	3m 11s	2026-09-25 12:24:01	Attack persisted - escalating to time ban
10.0.0.8	UDP Flood	0.5803	92.4%	High	Phase 2	1	1m 1s	2026-09-25 12:23:58	Ban Expired
10.0.0.20	UDP Flood	0.6452	79.4%	High	Phase 2	3	3m 1s	2026-09-25 12:23:51	Escalated to Time Ban
10.0.0.1	Uncertain	0.6111	58.9%	Low	Phase 1	1	10s	2026-09-25 12:23:24	Attack Stopped
10.0.0.1	Uncertain	0.6131	58.9%	Low	Phase 1	1	11s	2026-09-25 12:23:14	Attack Stopped
10.0.0.1	Uncertain	0.6136	58.9%	Low	Phase 1	1	11s	2026-09-25 12:23:03	Attack Stopped
10.0.0.17	SYN Flood	0.6466	92.2%	Critical	Phase 2	2	1m 11s	2026-09-25 12:22:01	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6452	79.4%	High	Phase 2	2	1m 1s	2026-09-25 12:21:51	Escalated to Time Ban
10.0.0.17	SYN Flood	0.6466	92.2%	Critical	Phase 2	1	11s	2026-09-25 12:21:01	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE